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1. Setup and configure a LAN using a switch and Ethernet cables

- **AIM:** Setup and Configure a LAN (Local area Network) using a switch and Ethernet cables in your lab
- what is a LAN ?

A local area network (LAN) is a computer network confined to a small geographic area, such as a home, office or computer lab. It links nearby devices like PCs, printers, and servers using Ethernet cables or wifi to enable fast data exchange and resource sharing.

- **Hardware Requirements :**

  => 4 Computers (PC1, PC2, PC3, PC4)

  => 1 Network Switch (8/16/24 ports)

  => 5 Ethernet cables

- **PROCEDURE :**
- Step 1 : physical connections

Connect each PC to a switch port using an Ethernet cable. Connect your management PC to the switch to configure it

- Step 2 : Configure IP Addresses on PCs

Navigate to control Panel → Network and Sharing Center → change adapter settings → Right-click Ethernet → properties → Internet Protocol Version 4 (TCP/IPv4)

Assign static IP addresses and Subnet Mask (255.0.0.0):

PC1 : 10.1.1.1

PC2 : 10.1.1.2

PC3 : 10.1.1.3

PC4 : 10.1.1.4

- Step 3: Configure the switch

open a web browser on your PC and enter the default switch IP. log in using administrator credentials. change the switch IP address to 10.1.1.5 with subnet mask 255.0.0.0

- Step 4: Test Network Connectivity

Open command prompt on PC1 and run:

successful replies confirm network connectivity

- Step 5: Configure File sharing

Create a folder on PC1, right-click it, select Properties rightarrow sharing tab.

click share...., select Everyone, and set permissions

On PC2, Press Win + R, type \\10.1.1.1 and press Enter to access shared files

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- RESULT: A local Area Network (LAN) was successfully set up, configured, and verified using ping command and cross network file sharing